

Lab B: Exp 21. The Wittig Reaction

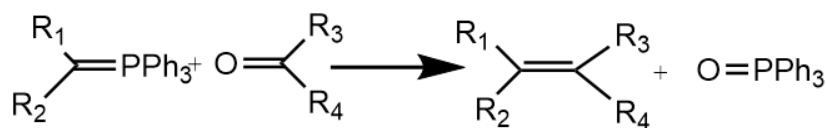
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I. INTRODUCTION

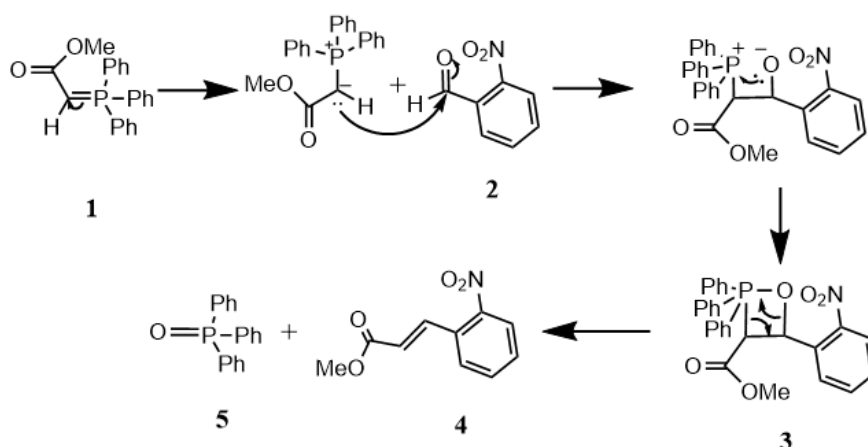
The Wittig reaction, is the reaction between an aldehyde or a ketone with a triphenyl phosphonium ylide, which is also called the Wittig reagent.¹ The Wittig reaction is an important reaction in organic chemistry as it yields the synthesis of an alkene¹, such that the location of the double bond is fixed, and thus predictable.² The Wittig reaction is named after George Wittig, who was awarded the Nobel prize in chemistry for this in 1979.³ The Wittig reaction is an important tool for synthesising diverse products such as pharmaceuticals, polymers, and cosmetics.⁴ Therefore, understanding the Wittig reaction mechanism provides insight into reactions that implement ylides, especially in modern medicinal chemistry and drug development. The general Wittig reaction is outlined in *Scheme 1* below.

Scheme 1: General overview of the Wittig reaction



The aim of this investigation is to synthesise and purify an alkene product from the Wittig reaction, to isolate the desired alkene using column chromatography, and to determine the purity and yield of the major alkene using thin-layer chromatography, and melting point analysis. In this investigation the Wittig reaction between 2-nitrobenzaldehyde and the ylide methyl (triphenylphosphoranylidene)acetate will be conducted. The exact reaction synthesis performed is depicted in *Scheme 2* below.

Scheme 2: The Wittig reaction mechanism



First, the ylide methyl (triphenylphosphoranylidene)acetate (1) is converted via resonance into its highly reactive carbanion. The lone pair on the carbon attacks the carbonyl carbon on 2-nitrobenzaldehyde (2). Consequently, the negatively charged oxygen forms a bond with the positively charged phosphor forming a ring intermediate called oxaphosphetane (3). The reaction proceeds as the oxaphosphetane intermediate (3) cleaves into the (E)-alkene product, methyl (E)-3-(2-nitrophenyl)acrylate (4) and triphenylphosphine oxide (5).

II. EXPERIMENTAL PROCEDURE

This experiment was conducted by Mark Rejna and Ibraheem Siddiqui following instructions outlined in the *Understanding the Principles of Organic Chemistry* textbook.⁵

Methyl (E)-3-(2-nitrophenyl)acrylate (4)

2-nitrobenzaldehyde (0.5 mmol, 75.6 mg), methyl (triphenyl phosphoranylidene)acetate (0.52 mmol, 176.0 mg), and silica gel (105 mg) were combined in a 1-dram vial. The solids in the vial were mixed thoroughly by twirling the spatula. The vial (uncapped) was then heated in a microwave oven at power 4 for 55 to 70 seconds. The reaction formed a sticky yellow liquid which solidified over time. Once the vial was heated and before the vial cooled down to room temperature, the vial contents were mixed using a spatula. After mixing, the solid formed a yellow powder. The resulting crude mixture was

purified via silica gel (0.6686 mg) column chromatography using 50:50 hexane:ethyl acetate as eluent. Fractions containing the target product, methyl (E)-3-(2-nitrophenyl)acrylate, were identified by TLC at R_f values of approximately 0.71 (*Table 1*) in 50:50 hexane:ethyl acetate mobile phase. Pure fractions were combined together (*Figure 1*). The product solution was evaporated under a stream of air with heating to yield methyl (E)-3-(2-nitrophenyl)acrylate (33.6 mg, 32.35 %). The evaporated product formed was a white powder. The melting range of methyl (E)-3-(2-nitrophenyl)acrylate was measured to be between 60.8 °C and 64.6 °C.

Triphenylphosphine oxide (5)

Fractions containing the byproduct, triphenylphosphine oxide, were identified by TLC at R_f values of approximately 0.1 (*Table 1*) in 50:50 hexane:ethyl acetate mobile phase. The byproduct was identified in lanes TO, C, 4, 5, 6, and 7 (*Figure 1*). Byproduct in later fractions indicates the completion of the reaction and verifies a successful separation of the reaction components.

III. DISCUSSION

RESULTS AND DATA ANALYSIS

The mass of the target product, methyl (E)-3-(2-nitrophenyl)acrylate, recovered was 0.0336 g. Fractions were evaluated using TLC to identify their contents and to isolate the target product. As seen in *Figure 1* and *Table 1*, triphenylphosphine oxide was identified with an R_f value of approximately 0.1. This byproduct was present in the crude mixture and fractions 4 through 7. The target product, methyl (E)-3-(2-nitrophenyl)acrylate, was identified by a distinct spot with an R_f value of approximately 0.71, this was first seen in the crude mixtures. Fractions 1, 2, and 3 contained pure target product. Later fractions of 4 through 7 contained triphenylphosphine oxide, thus indicating the completion of the reaction and verifying a successful separation of the reaction components.

Figure 1: TLC plate results

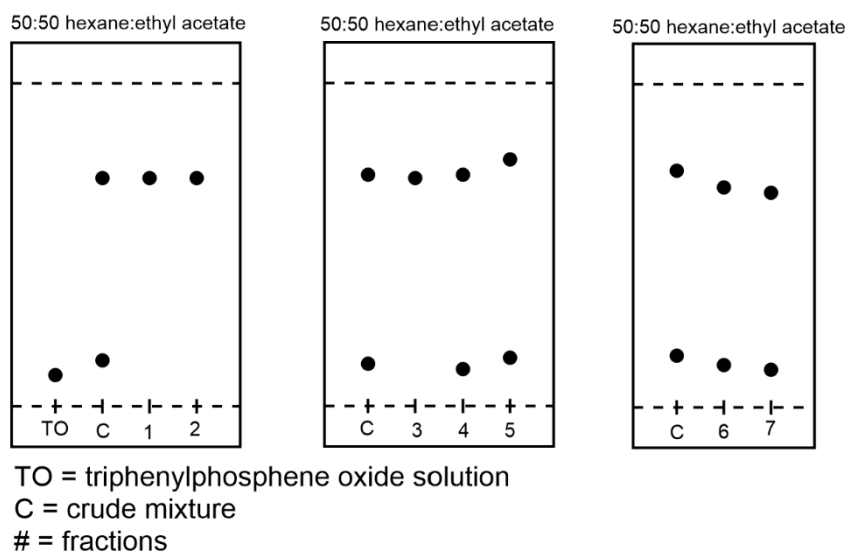


Table 1: TLC plate results – R_f values

Label	Rf value	
	Spot 1	Spot 2
TO	0.098	/
C (plate 1)	0.137	0.706
1	/	0.706
2	/	0.706
C (plate 2)	0.127	0.718
3	/	0.709
4	0.109	0.724
5	0.145	0.764
C (plate 3)	0.163	0.731
6	0.131	0.683
7	0.115	0.673

The percent yield of this experiment was calculated to be 32.35 %. This yield is lower than expected, one reason for this is that a product was lost during the transfer between the vial and the column. To prevent this error, more attention and caution should have been given and better solution wash practices could be implemented. Another potential cause for this low yield is that a significant amount of purified product was lost during the evaporation process

in the fume hood, where the stream of air was initially set too high causing some of the product solution to spill out. This human error could be minimised by starting the air stream at a low speed first and then placing the filter flask containing product under the more stable air stream. An alternative cause for the low yield is due to band broadening, where the target product is spread over many fractions. This error was due to improperly adding solvent to the column causing an overloaded column. To resolve this error, a longer column could be used, this would give a better separation and in turn yield better fractions, however, the column chromatography process would take more time. Another solution to this error is to collect more fractions, each of less volume, for example, 0.5 ml instead of 1 ml, to prevent the loss of yield in mixed fractions.

The melting range of the target product, methyl (E)-3-(2-nitrophenyl)acrylate, was measured to be between 60.8 °C and 64.6 °C. The literature melting point for methyl (E)-3-(2-nitrophenyl)acrylate is 72 °C.⁶ The measured melting point is less than the literature melting point, this indicates that the yielded product contains impurities. The most likely impurity is the byproduct, triphenylphosphine oxide, which must have occurred in trace amounts as the TLC analysis did not detect it. Alternative impurities may include unseparated crude mixture or unreacted starting materials that remained in the sample. To remove this error, recrystallisation could have been implemented to better purify the sample or alternatively gradient elution could have been used instead of TLC.

CONCLUSION

Overall, the Wittig reaction performed in this investigation was successful as the target product, methyl (E)-3-(2-nitrophenyl)acrylate, was identified using TLC and confirmed using melting point analysis. The column chromatography effectively separated the target product and byproduct, triphenylphosphine oxide, these had distinct spots at R_f values of

approximately 0.1 and 0.71. The measured melting range of 60.8 °C - 64.6 °C was less than the literature value of 72 °C, thus indicating the trace presence of impurities. The yield of 32.35 % was lower than expected. With implementing the improvements of the errors listed above, the purity and yield can be improved.

IV. FURTHER DISCUSSION

To improve this experiment, increase percent yield and purity of the target product, transfer loss between vial and column should be minimised, loss should be avoided during evaporation, and column overloading should be avoided. Additionally, if time permitted, a longer column could be used to better separate the products and more fractions can be collected for more precision. Finally, recrystallisation could be implemented to increase the purity of the target product.

V. REFERENCES

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